Anushka Murthy

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Education

Stanford University *2022 – Present*
PhD in Mathematics (2022-2023), Operations Research (2023-present) *Stanford, CA*

- Advisors: Ramesh Johari and Irene Lo.
- Awards: Stanford Human-Centered AI and Data Science Scholarship, NSF Graduate Fellowship, EDGE Fellowship.
- Research focus on causal inference and market design in stochastic systems
- Selected coursework: Graduate Analysis and Algebra, Stochastic Systems, Optimization, Causal Inference, Topics in Social Data, Game Theory, Topics in Market Design.
- Activities: Dean's Graduate Student Advisory Council, Stanford Women in Math Mentoring, mentor in Undergraduate Diversity in Research program, violinist in music department.

Columbia University School of Engineering and Applied Science *2018 – 2022*
Bachelor of Science in Applied Math *New York, NY*

- Graduated *summa cum laude*, GPA: 4.07/4.33.

Papers and Interests

Interests: My methodological interests are in applied probability, market design, and causal inference. I am particularly interested in the impact of data-driven interventions (A/B experiments, recommender systems, AI agents) on information acquisition and decision-making in marketplaces.

1. *When Does Interference Matter? Decision-Making in Platform Experiments*, R. Johari, H. Li, A. Murthy, G. Weintraub. Major revision at **Management Science**.
2. *Dynamic Contracting for Payment for Ecosystem Services*, I. Lo and A. Murthy. Major revision at **Management Science**.
3. *One-point asymptotics for half-flat ASEP*, E. Dimitrov and A. Murthy. **Annals of Applied Probability**, 34(1B): 1136-1176 (2024).

Industry Experience

Amazon Middle Mile Marketplace Science *June – September 2025*
Applied Scientist Intern *Bellevue, WA*

- Developed machine learning models to predict trailer pool adjustment cost in the spot market.
- Validated and enhanced estimation methods for quantifying the causal impact of increasing lane volume through Amazon Freight contracts on trailer pool adjustment costs, enabling more accurate cost forecasting.

Teaching Experience

Simons Laufer Mathematical Sciences Institute (MSRI) *June 2023*
Teaching Assistant *Berkeley, CA*

- Assisted in graduate summer school on "Mathematics and Computer Science of Market Design."

Stanford University *2022 – Present*
Course Assistant *Stanford, CA*

- Courses Taught: Math 18 (Foundations for Calculus), Math 113 (Linear Algebra), MS&E 232 (Game Theory), MS&E 232H (Accelerated Game Theory), OIT 272 (Online Marketplaces, MBA Elective).

Columbia University *2020 – 2022*
Teaching Assistant *New York, NY*

- Courses Taught: COMS 3261 (Computer Science Theory), MATH GU4061 (Modern Analysis I), MATH GU4032 (Fourier Analysis), MATH UN3028 (Partial Differential Equations), MATH UN2010 (Linear Algebra), MATH UN1205 (Accelerated Multivariable Calculus).

One-to-One Tutoring, Columbia University *2020 – 2022*
Tutor *New York, NY*

- Math tutor for elementary school students, focusing on improving skills and study habits.

Awards and Recognition

Lillie Fund (2026)

Stanford Human-Centered AI and Data Science Scholarship (2024-2026)

NSF Graduate Research Fellowship (2022-2027)

Stanford EDGE Doctoral Fellowship (March 2022)

Columbia Provost's Diversity Fellowship (March 2022)

UC Berkeley Chancellor's Fellowship (February 2022)

Dean's List (All semesters at Columbia).

Talks

"When Does Interference Matter? Decision-Making in Platform Experiments."

Economics and Computation 2026, Rome — July 2026
CODE at MIT 2025 — November 2025
INFORMS 2025, Atlanta — October 2025
Experimentation in Operations Workshop, Harvard Business School — May 2025
Data-Driven Decisions Seminar, Stanford GSB — May 2025

"False Positives in A/A Experiments with Correlated Outcomes."

INFORMS 2024, Seattle — October 2024
CODE at MIT 2024 — October 2024
Stanford Causal Science Center Conference — October 2024

"Dynamic Contracting for Payment for Ecosystem Services Programs."

MSOM 2024, Minneapolis — July 2024

Skills

Languages: Python, Julia, R, MATLAB, Java, C++.

Libraries/Technologies: NumPy, Matplotlib, Scikit-learn, cvxpy, Pandas.

References

Ramesh Johari — *Stanford University*
Professor of Management Science and Engineering

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Irene Lo — *Stanford University*
Assistant Professor of Management Science and Engineering

ilo@stanford.edu

Gabriel Weintraub — *University of Chile/Columbia Business School*
Professor of Economics and Business

gweintra1@gmail.com